Proposed dual-radio-map-aware IGA-TD3 framework

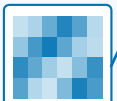
Heterogeneous inspection payloads | Joint sequence and continuous-trajectory optimization

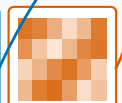
System inputs

Fixed inspection scenario

Heterogeneous payloads
D = [D1, ..., DK]

Fixed dual radio maps


G2A outage map


A2G uplink-SNR map

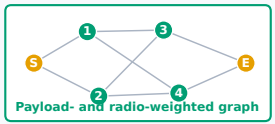
Shared mission constraints

- Fixed start/end and inspection points
- Motion and boundary limits
- QoS budget T_out <= T_out^max

Dual-map-aware IGA

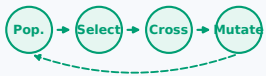
Edge feature construction

Geometric distance	d_ij
Source payload	D_i
Expected outage duration	tau_out,ij
Expected uplink throughput	R_ij



Payload- and radio-weighted graph

w_ij = f(d_ij, D_i, tau_out,ij, R_ij)
Fitness F(O) = sum w_ij



Optimized inspection order O*

Map-aware TD3

UAV inspection environment

Move - map query - upload

reward r_n, next state s_(n+1)

action a_n

Actor
pi_phi(s_n)

Training only
MapGuide feasible demonstrations
BC anchor

Map-aware state s_n
[q, target - q, D_buf, T_out/T_max, A2G SNR, G2A outage probability]

Continuous action a_n
relative heading offset + flight distance

delayed soft updates

policy

Replay buffer
~~(s, a, r, s')~~

Twin critics
Q1(s,a) Q2(s,a)

Target nets
min target Q

Guide-free Actor (evaluation)

Mission output

Continuous trajectory Q*

- ✓ All points visited
- ✓ All payloads uploaded
- ✓ Destination reached

Optimization objective
min T + tau_total

Hard QoS constraint
T_out <= T_out^max

T, tau_total, T_out, flight distance

Information flow | Training/update | G2A | A2G